Executive Summary

FinTrust Bank: Credit Risk & Retention Analytics (prototype)

What we were asked

Reduce losses from loans that are not repaid, and stop valuable customers from leaving. We built a working prototype that (1) rates loan applicants by default risk, (2) flags existing customers likely to leave, and (3) groups customers into segments with a suggested action for each. A dashboard lets the risk and retention teams use it.

1. Loan default risk

We studied 150,000 past borrowers; about 1 in 15 (6.7%) ran into serious repayment trouble. Our model sorts applicants into three bands. On applicants the model had never seen, the bands performed as follows:

Risk band	Applicants tested	Actually defaulted
Low (under 5% estimated risk)	20,926	1.7%
Medium (5-20%)	5,975	9.6%
High (over 20%)	2,946	36.6%

The riskiest 10% of applicants account for 54% of all defaults. **What drives risk:** a history of late payments (especially 90+ days late) and using nearly all of the available credit limit. Younger applicants also default more often (11.9% aged 18-29 vs 2.3% aged 70+). Income matters much less than payment behaviour.

2. Customer churn (customers leaving)

About 1 in 5 customers (20.4%) in the 10,000-customer file left the bank. The model ranks customers by their likelihood of leaving: the 1,407 customers in the High band (14%) actually left 74.1% of the time and contain 51% of all leavers, while the Low band left only 5.8% of the time. Patterns we found: customers aged 50-59 leave at 56%; German customers at 32% (about double France and Spain); inactive members at 27% (vs 14% for active); customers with exactly two products leave least (7.6%).

3. Customer segments

Segment	Customers	Left the bank	Suggested action
High-balance, inactive	1,979	32.4%	Top priority: personal outreach
Senior customers	961	32.0%	Relationship manager; retirement products
Multi-product, inactive	1,102	31.4%	Review product fit and fees
Dormant, near-zero balance	1,728	16.4%	Low-cost digital reminders
Active, high-balance	2,505	12.6%	Protect and cross-sell
Active, low-balance	1,725	8.2%	Loyal base; help balances grow

Recommended actions this quarter

1. Lending: use the three bands to prioritise review. Fast-track Low-band applicants; send High-band applicants for manual review or tighter terms. Use the model to support, not replace, credit officers.
2. Retention: focus the retention budget on the three segments with roughly 32% churn (about 4,000 customers) rather than all customers, starting with the High-balance inactive group.
3. Investigate: why German customers leave twice as often, and why customers with 3-4 products leave so often (small group, 326 customers).
4. Test before scaling: run the suggested retention actions on a sample and measure whether churn actually falls.

Things to keep in mind

The loan data and the customer data describe different people, so we report them separately; we cannot say how likely one individual is to both default and leave. About 6 in 10 applicants the loan model flags as likely defaulters do not actually default, so it is a screening aid, not an automatic decision. Segments overlap rather than being sharply separate. Age, gender and country influence the models, so a fairness review is needed before real use. Suggested actions are recommendations, not proven results.